Design and Performance Analysis of Multirate-NOMA

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CERTIFICATE

This is to certify that the project **entitled "Design and Performance Analysis of Multirate-NOMA"** is the bonafide work done by SUKRUTHA BODAPALLY (23R21A0472), , ALEKHYA (23R21A0485), BALAJI (23R21A0492) in partial fulfilment of the requirement for the award of the degree of B.Tech in Electronics and Communication Engineering, during the academic year 2026-27.

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ABSTRACT

Non-Orthogonal Multiple Access (NOMA) has emerged as a promising multiple access technique for improving spectral efficiency by enabling multiple users to share the same time-frequency resources through power-domain multiplexing. However, most existing studies focus on conventional Single-Rate (SR)-NOMA, where only the transmit power allocated to each user is varied. This limitation reduces system flexibility and restricts the achievable Bit Error Rate (BER) performance. To address these challenges, this paper investigates the design and performance analysis of a Multi-Rate (MR)-NOMA framework, which jointly adjusts the symbol rate and transmit power to control the symbol energy allocated to each user. A Joint Multiuser Maximum Likelihood Sequence Detector (JMLSD) based on the Maximum Likelihood (ML) criterion and a novel low-complexity optimal Successive Interference Cancellation (SIC) receiver are developed to efficiently detect multiuser signals. In addition, closed-form BER expressions are derived for arbitrary symbol rates and modulation orders, providing a comprehensive analytical performance evaluation. The derived BER formulations are further employed to optimize power allocation at the Base Station (BS) while satisfying predefined BER constraints for all users. Simulation results validate the analytical expressions and demonstrate that MR-NOMA provides greater flexibility in balancing spectral efficiency and communication reliability. Compared with conventional SR-NOMA, the proposed MR-NOMA scheme achieves significantly improved BER performance, with gains of up to two orders of magnitude in challenging communication scenarios.

TABLE OF CONTENTS

Certificate ii
Declaration iii
Acknowledgement iv
Abstract v
List of Figures vi
List of Tables vii

CHAPTER NO **PG.NO.**

CHAPTER 1: INTRODUCTION **1-5**
1.1 Introduction 1
1.2 Background of the Project 1
1.3 Problem Statement 3
1.4 Objectives of the Project 3
1.5 Scope of the Project 4
CHAPTER 2: LITERATURE SURVEY **6-8**
2.1 Introduction 6
2.2 Review of Existing Systems 6
2.3 Literature Survey of Related Works 7
2.4 Limitations of Existing System 8
2.5 Research Gap 8
2.6 propososed system overiew 8
CHAPTER 3: SYSTEM DESIGN AND METHODOLOGY **9-15**
3.1 Introduction 9
3.2 System Architecture / Block Diagram 9
3.3 Software Requirements 20
3.4 System Design and Working Principle 11
3.5 Algorithm / Flowchart 13
3.6 Implementation Methodology 14
CHAPTER 4: RESULTS AND DISCUSSION **16-25**
4.1 Introduction 16
4.2 Experimental Setup 16
4.3 Implementation Details 17

4.4 Results and Observations 17
4.5 Performance Analysis 22
4.6 Comparison with Existing Systems 23
4.7 Discussion of Results 24
CHAPTER 5: CONCLUSION AND FUTURE SCOPE 26-27
5.1 Conclusion 26
5.2 Key Contributions and Outcomes 26
5.3 Limitations of the Project 27
5.4 Future Scope 27
REFERENCES 28-29
APPENDICES 29-31

LIST OF FIGURES

FIG.NO	DESCRIPTION	PAGE NO.
Fig.1.1	Concept of orthogonal multiple access and power-domain NOMA	2
Fig.2.1	Classification of multiple-access techniques	5
Fig.3.1	MR-NOMA base-station block diagram	8
Fig.3.2	Proposed MR-SIC detector block diagram	8
Fig.3.3	Constellation of U_2 after sample combining	11
Fig.3.4	Constellation of U_2 after sample combining	12
Fig.3.5	Flowchart of the MR-SIC detection algorithm	13
Fig.3.6	Flowchart of the analytical JMLSD BER evaluation program	14
Fig.4.1	BER performance of MR-NOMA, JMLSD versus SIC	19
Fig.4.2	Preliminary MATLAB output from Review-2: (a) BER versus SNR, (b) BER versus α_1	19
Fig.4.3	BER of the JMLSD and SIC detectors from the base paper	21

LIST OF TABLES

TABLE.NO	DESCRIPTION	PAGE NO.
Table 1.1	Objectives of the proposed project	1
Table 1.2	Applications of Multirate-NOMA	3
Table 2.1	Summary of related literature	6
Table 2.2	Comparison of existing multiple-access systems	7
Table 3.2	Software requirements	9
Table 3.4	System parameters used in the project	10
Table 3.5	Computational complexity of JMLSD and MR-SIC (from the base paper)	11
Table 3.6	MATLAB files and their roles	15
Table 3.7	Project timeline	15
Table 4.1	Experimental setup	17
Table 4.2	Mapping of paper quantities to MATLAB implementation	18
Table 4.3	BER values from the MATLAB result file	18
Table 4.4	Comparison of the result file with the base paper (approximate readings)	21
Table 4.5	Gains of MR-NOMA over SR-NOMA reported in the base paper	21
Table 4.6	Advantages and limitations of MR-NOMA	22
Table 5.1	Key contributions and outcomes	23

CHAPTER 1: INTRODUCTION

1.1 Introduction

The growth in the number of connected devices and in the demand for high data rates requires wireless systems that use the available spectrum more efficiently. Non-orthogonal multiple access (NOMA) is a promising technique for this purpose because it improves spectral efficiency, supports massive connectivity and allows users with different quality-of-service (QoS) requirements to be served at the same time [1], [2]. In power-domain NOMA the base station (BS) superimposes the signals of several users in the same time and frequency resource and separates them by assigning different power coefficients, while the resulting multiuser interference is removed at the receivers by successive interference cancellation (SIC) [3], [4].

Most published analyses consider single-rate NOMA (SR-NOMA), in which all users have the same symbol rate and only the power coefficient can be varied per user. This limits both the design flexibility and the BER. Multirate NOMA (MR-NOMA) removes this limitation by giving each user its own symbol rate, so that the symbol energy of a user depends on both its power coefficient and its symbol duration [1], [5]. This project studies the design and BER performance of a downlink MR-NOMA system based on [1] and presents the BER results obtained from our MATLAB implementation.

1.2 Background of the Project

In a two-user downlink NOMA system the BS transmits $x = \sqrt{\alpha_1}x_1 + \sqrt{\alpha_2}x_2$, where the user closest to the BS (U_1) receives the smaller power coefficient α_1 and the farthest user (U_2) receives the larger coefficient α_2, with $\alpha_1 + \alpha_2 = 1$. The far user suffers from stronger path loss and from interference by the near user, so it is given more power. The near user first detects and removes the far user's signal using SIC and then detects its own signal [3], [6].

In MR-NOMA the symbol durations $T_1 < T_2 < \ldots < T_N$ are also different. The far, delay-tolerant user is given a long symbol duration, which makes its detection more reliable because more signal energy is collected per symbol, whereas the near user keeps the shortest

symbol duration. Fig. 1.1 illustrates this for two users with $T_2 = 2T_1$: the far user's symbol is held over two samples while the near user's symbol changes at every sample. The constraint $T_N/T_n = \tau_n \in \mathbb{Z}$ allows all receivers to use one common sampling rate [1].

Table 1.1: Evolution of multiple-access schemes

Scheme	**Domain**	**User access mechanism**	**Examples**	**Notes**
FDMA	Frequency	Dedicated frequency per user	1G analog cellular	Simple, continuous transmission
TDMA	Time	Dedicated time slot per user	GSM, T1/E1	Cyclical bursts, needs synchronisation
CDMA	Code	Unique code per user	3G W-CDMA, GPS	Spread spectrum, high concurrency
OFDMA	Frequency /Time	Subcarrier blocks per user	LTE, WiMAX	Orthogonal subcarriers, high efficiency
NOMA	Power/Code	Power-division (or code-based) multiplexing	5G NR	Non-orthogonal, requires SIC

In power-domain NOMA the base station superimposes the signals of several users on the same resource and gives the far user the larger share of power, as illustrated in Fig. 1.1. The near user first detects and removes the far-user signal using SIC and then detects its own signal. This improves spectral efficiency, supports massive connectivity and lets users with different QoS requirements be served together.

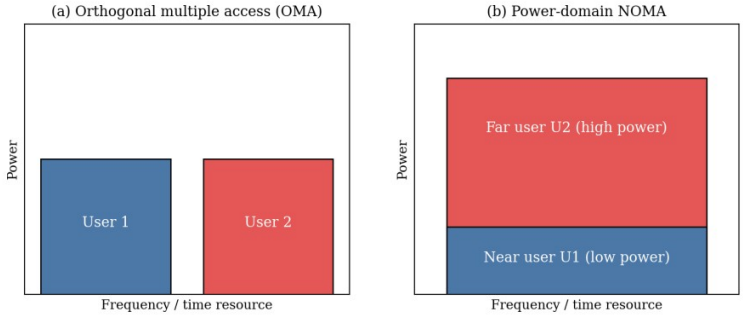


Fig. 1.1: Concept of orthogonal multiple access and power-domain NOMA

The main drawback of NOMA is the error rate degradation caused by multiuser interference. In single-rate NOMA the only tunable quantities are the power allocation and the modulation order. MR-NOMA, proposed for single-carrier systems, adds the symbol rate as a third degree of freedom. A far user can be given a longer symbol, which raises its energy per symbol and improves detection, while a near user keeps a short symbol and a high rate. This matches the mixed traffic expected in 6G, where delay-sensitive vehicle-to-vehicle links and delay-tolerant sensors share the same cell.

1.3 Problem Statement

In SR-NOMA the BER of the users is determined by the modulation orders and the power allocation only, and the multiuser interference cannot be reduced without increasing the power of the far user at the expense of the near user. For MR-NOMA, numerical BER results have been reported without analytical derivations [9], [5]. The problem addressed in this project is therefore to understand how the use of different symbol rates changes the detection process and the BER of a NOMA system, to implement a low-complexity receiver for it, and to quantify the BER improvement over SR-NOMA for different modulation orders and symbol-duration ratios.

1.4 Objectives of the Project

The aim of the project is to study, model and evaluate a two-user downlink MR-NOMA system that jointly controls symbol rate and transmit power, and to validate its analytical BER performance against conventional SR-NOMA.

Table 1.2: Objectives of the proposed project

Sl. No.	Objective	Status in this project
1	Design a downlink MR-NOMA system with arbitrary modulation orders and symbol rates	Two-user model with T = [1, 2] formulated
2	Study the optimal JMLSD and its BER upper bound (PEP and union bound)	JMLSD analytical BER implemented in MATLAB
3	Study the low-complexity MR-SIC receiver	Far-user and near-user MR-SIC BER implemented
4	Analyse power allocation that minimises one user's BER while meeting the other's BER requirement	Studied; BER-versus-α_1 sweep in progress
5	Quantify the BER gain of MR-NOMA over SR-NOMA ($[T_1, T_2] = [1, 1]$)	Planned with Monte Carlo validation
6	Explore extension to multi-antenna NOMA	Future scope

1.5 Scope of the Project

The project covers a downlink system with one base station and two users (near and far) over an AWGN channel, with BPSK modulation, symbol durations T = [1, 2] and the equal-distance power allocation α = [0.2, 0.8]. The analysis is analytical and implemented in MATLAB. Monte Carlo simulation, higher-order QAM, larger T_2, optimisation of power allocation, fading channels, uplink operation and more than two users are outside Stage 1 and are listed as future scope. Table 1.3 lists the application areas for which MR-NOMA is intended.

Table 1.3: Applications of Multirate-NOMA

Application area	**How MR-NOMA helps**
6G eMBB, uRLLC and mMTC coexistence	Short symbols for latency-critical users and long symbols for delay-tolerant devices on one resource
Vehicle-to-vehicle and vehicular networks	Low-latency safety messages served beside bulk traffic
Massive IoT and sensor networks	Far, low-power devices that tolerate delay gain reliability from longer symbols
Satellite-terrestrial, aerial and RIS-aided links	Power-domain and rate-domain freedom can be combined with these architectures
Cell-edge coverage improvement	The far user receives more symbol energy without a very large power share

CHAPTER 2: LITERATURE SURVEY

2.1 Introduction

The literature around MR-NOMA falls into three families: multinumerology NOMA, partial NOMA, and multirate NOMA itself. Five recent papers were selected for Review-1, and the related-work section of the base paper was used to place them in context. Fig. 2.1 shows how these techniques are classified.

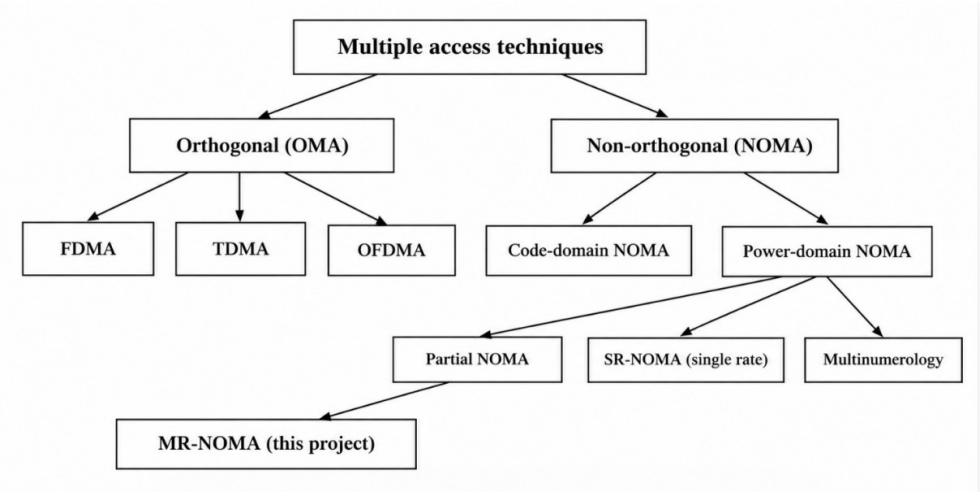


Fig. 2.1: Classification of multiple-access techniques

2.2 Review of Existing Systems

Single-rate NOMA. All users have the same symbol rate. The BER of SR-NOMA with an arbitrary number of users and modulation orders has been derived in closed form [6], the power allocation for reliable SIC detection of rectangular QAM has been studied [10], and NOMA with adaptive modulation and power under block-error-rate constraints has been analysed [11]. A survey of the error-rate analysis of NOMA is given in [3].

Multinumerology NOMA. Different subcarrier spacings, and hence different symbol durations, are used for OFDM-based users and are multiplexed in the power domain, which leads to severe inter-numerology interference [1]. In [7] the uplink two-user case with SIC was analysed, and in [12] the work was extended to an arbitrary number of users with joint subcarrier and power allocation. A downlink two-user system with time-domain

superposition was studied in [13].

Partial NOMA. The subcarrier spacing is the same for all users, but the extent of the frequency overlap between users is controlled. For two users, expressions for the signal-to-interference-plus-noise ratio and the sum rate were derived in [8], showing a better sum rate than OMA or fully overlapped NOMA, and the scheme was studied in Poisson networks in [14].

Multirate NOMA. MR-NOMA is proposed for single-carrier systems. Users are assigned different symbol rates according to their requirements, so that they experience different overlaps in time and frequency [1]. Uplink BER results were presented numerically, without analytical derivations, with multilevel irregular repeat-accumulate channel coding in [9] and for an uncoded system with arbitrary symbol rates in [5].

2.3 Literature Survey of Related Works

Table 2.1: Summary of Related Literature

Ref.	Scheme	Main contribution
[3]	SR-NOMA	Survey of error-rate analysis of NOMA: principles and future directions.
[6]	SR-NOMA	Exact BER analysis for an arbitrary number of users and modulation orders.
[10]	SR-NOMA	Power allocation for reliable SIC detection of rectangular QAM.
[11]	SR-NOMA	Adaptive modulation and power under BLER constraints.
[7]	Multinumerology	Uplink two-user inter-numerology interference with SIC; exhaustive search for the power allocation that maximises spectral efficiency.
[12]	Multinumerology	Generalisation to an arbitrary number of users; subcarrier and power allocation for spectral efficiency.

2.4 Limitations of Existing Systems

SR-NOMA has only two degrees of freedom, the modulation orders and the power allocation, and the interference from the near user limits the BER of the far user [1]. Multinumerology and partial NOMA require multicarrier processing and suffer from inter-numerology interference or from the complexity of controlling the frequency overlap [7], [8]. The existing MR-NOMA studies report numerical BER results only and do not derive the detectors or the BER analytically [9], [5].

2.5 Research Gap

There was no detector design and closed-form BER analysis for downlink MR-NOMA that covers arbitrary symbol rates and modulation orders. The base paper fills this gap by designing the optimal JMLSD and a low-complexity MR-SIC receiver, deriving the BER expressions and using them for power allocation [1]. The gap addressed by this project is the implementation and verification of the far-user BER behaviour of this scheme for several modulation pairs and symbol-duration ratios, and its comparison with the published results.

2.6 Proposed System Overview

The system studied in this project is a two-user downlink MR-NOMA system. The BS superimposes the symbols of a near user with symbol duration $T_1 = 1$ and a far user with symbol duration $T_2 \in \{1, 2, 4, 8\}$. The far user combines the T_2 samples of each of its symbols and detects directly, while the near user uses the MR-SIC receiver, which first detects the far user's symbol, removes it, and then detects its own symbols [1]. The far-user BER of the system is evaluated in MATLAB for different modulation pairs and compared with SR-NOMA ($T_2 = 1$) and with the base paper.

CHAPTER 3: SYSTEM DESIGN AND METHODOLOGY

3.1 Introduction

This chapter describes the MR-NOMA signal model, the optimal JMLSD and the low-complexity MR-SIC receiver, their computational complexity, and the methodology used to evaluate the BER. The model and expressions follow [1]; the notation of [1] is retained.

3.2 System Architecture / Block Diagram

A single base station serves NN users in the downlink, with this project considering N=2. User U1 is the near user with a higher symbol rate, while U2 is the far user with a lower symbol rate, so T1<T2. The ratio $t2/t1 = \tau$ must be an integer to maintain a common sampling rate. Two independent modulators generate symbols at intervals T1and T2, respectively. Their outputs are scaled by $x[\ell] = \sqrt{\alpha_1}x_1[\ell] + \sqrt{\alpha_2}x_2[\ell]$, then combined to form the transmitted signal $x1[\ell]$and$x2[\ell]$. The signal reaches users U1 and U2 through their respective channels, h1and h2 [1].

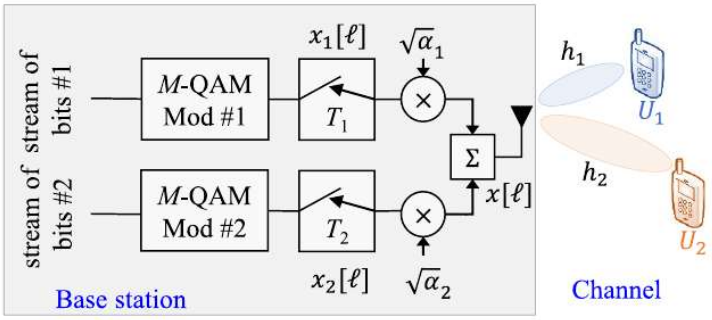


Fig. 3.1: MR-NOMA base-station block diagram for N = 2

At the receivers (Fig. 3.2), U_2 samples at rate $1/T_1$, sums T_2 samples and demodulates its own symbol. U_1 first detects x_2 in the same way, remodulates and scales it by $\sqrt{\alpha_2}$, subtracts it from the received samples and then demodulates x_1.

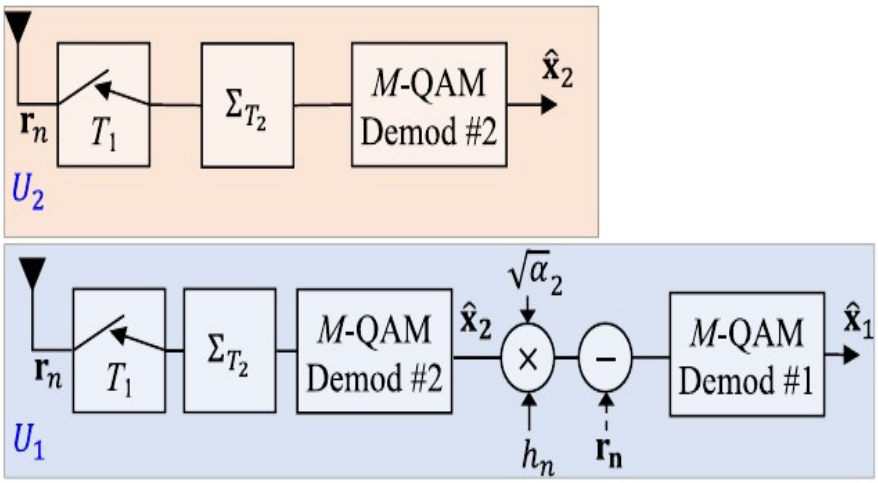


Fig. 3.2: Proposed MR-SIC detector block diagram for N =2

3.3 Software Requirements

Table 3.1: Software requirements

Software	Version / detail	Purpose
Operating system	Windows (used in the ECE laboratory)	Platform for MATLAB
MATLAB	R2007b (version 7.5) or later	Implementation, BER computation and plotting
MATLAB Publish	Built in	Produces the HTML/PDF result report
Presentation and document software	PowerPoint, Word	Review presentations and this report
PDF viewer	Any	Viewing the base paper and result files

Table 3.2 explains the software tools used in the MR-NOMA project. The operating system is Windows, which provides the environment for running the required applications.

MATLAB Publish, a built-in feature, is used to generate reports containing the MATLAB

code and simulation results. PowerPoint and Word are used to prepare the project presentation documentation. Finally, a PDF viewer is used to read the base research paper and view the simulation result files.

3.4 System Design and Working Principle

Signal model. Consider N users with normalised symbol durations $T_1 < T_2 < \dots < T_N$, $T_1 = 1$, where U_1 is nearest to the BS and has the highest symbol rate and U_N is the farthest with the lowest symbol rate, and $T_N/T_n = \tau_n$ is an integer. The power coefficients satisfy $\Sigma\alpha_n = 1$ and $\alpha_1 < \alpha_2 < \dots < \alpha_N$, which allows reliable detection at the users [1], [10]. The transmitted samples over a window of T_N sampling periods are

$$x[\ell] = \sum_{n=1}^{N} \sqrt{\alpha_n}\, x_n[\ell], \quad \ell \in \{1,2, ..., T_N\} \tag{3.1}$$

or, in vector form, with x_n containing τ_n different symbols spread uniformly over the window,

$$\mathbf{x} = \sum_{n=1}^{N} \sqrt{\alpha_n}\, \mathbf{x}_n \tag{3.2}$$

The bit rate of user n, with M_n the constellation size, is

$$R_n = \frac{\log_2 M_n}{T_n} \tag{3.3}$$

Over the same window, the signal received by the nth user is

$$\mathbf{r}_n = h_n \mathbf{x} + \mathbf{w}_n, \mathbf{w}_n \sim \mathrm{CN}\left(0, N_0 \mathbf{I}_{T_N}\right) \tag{3.4}$$

where h_n is the channel gain, w_n is the AWGN vector and N_0 is the noise power spectral density. For N = 2 and $T_2 = 2$, for example, $x[1] = \sqrt{\alpha_1} x_1[1] + \sqrt{\alpha_2} x_2[1]$ and $x[2] = \sqrt{\alpha_1} x_1[2] + \sqrt{\alpha_2} x_2[1]$, because the far-user symbol is repeated over both samples [1].

Optimal detector (JMLSD). For a given h_n and trial sequence $\tilde{x}$, the received vector follows a multivariate complex Gaussian density

$$f(\mathbf{r}_n; h_n, \mathbf{x} = \tilde{\mathbf{x}}) = \frac{1}{(\pi N_0)^{T_N}} \exp\left(-\frac{\|\mathbf{r}_n - h_n \tilde{\mathbf{x}}\|_2^2}{N_0}\right) \tag{3.5}$$

and maximising it gives the joint multiuser maximum-likelihood sequence detector

$$\{\hat{\mathbf{x}}_1, ..., \hat{\mathbf{x}}_N\} = \underset{\tilde{\mathbf{x}}}{\operatorname{argmin}} \|\mathbf{r}_n - h_n \tilde{\mathbf{x}}\|_2^2 \tag{3.6}$$

When all users have the same symbol rate, the JMLSD reduces to the symbol-by-symbol joint multiuser maximum-likelihood detector [1].

Low-complexity MR-SIC detector. Since $\alpha_N > \alpha_n$ and $T_N > T_n$ for all $n < N$, the detection order is $x_N, x_{N-1}, \ldots, x_1$. The detector at user n is a generalisation of the SR-SIC detector and works in the following steps [1]. In Step 1 the T_N received samples are summed and the symbol of the lowest-rate user is detected with a single-user MLD,

$$\hat{x}_N = \arg\min_{\tilde{x}_N} \left| \mathbf{1}_{T_N}^{\top} \mathbf{r}_n - h_n \sqrt{\alpha_N} T_N \tilde{x}_N \right|^2 \tag{3.7}$$

where $1_{TN}{}^{T} r_n$ is the sum of the T_N received samples. In Step 2, for $n < N$, the interference of the higher-energy users that have already been detected is removed,

$$\tilde{\mathbf{r}}_n = \mathbf{r}_n - h_n \sum_{i=n+1}^{N} \sqrt{\alpha_i}\, \hat{\mathbf{x}}_i \tag{3.8}$$

where each detected symbol is repeated over the samples it occupies. In Step 3 the vector $\tilde{r}_n$ is partitioned into τ_n equal vectors $\tilde{r}_n{}^1, \ldots, \tilde{r}_n{}^{\tau_n}$ of length T_n. In Step 4 each symbol of user n is detected with a single-user MLD,

$$\hat{x}_n[j] = \arg\min_{\tilde{x}_n} \left| \mathbf{1}_{T_n}^{\top} \tilde{\mathbf{r}}_n^{j} - h_n \sqrt{\alpha_n} T_n \tilde{x}_n \right|^2, j = 1, \ldots, \tau_n \tag{3.9}$$

For $T_n = 1$ for all n, the detector reduces to the conventional SR-SIC detector. The BS must provide α_i, T_i and M_i of all higher-energy users $i \geq n$ to user n, which is similar to SR-NOMA [1].

Equivalence with the JMLSD. For the low-rate user, Appendix A of [1] shows that the JMLSD metric, after expansion, contains the terms $T_2\alpha_2|h_n|^2|\tilde{x}_2|^2 - 2\sqrt{\alpha_2}\mathrm{Re}\{(\Sigma r_n{}^*[\ell])h_n\tilde{x}_2\}$, while the metric of the combining detector in Eq. (3.7) has the same two terms multiplied by T_2. Dividing by the common factor T_2 makes them identical, so both detectors decide in the same way, and the subsequent detection of x_1 is then also equivalent. The MR-SIC receiver is therefore an optimal low-complexity implementation of the JMLSD [1].

Complexity. The JMLSD must search over all trial sequences, whose number is $\prod M_i{}^{\tau_i}$. The numbers of real multiplications (RM) and real additions (RA) derived in [1] are

$$\mathrm{CC}_{\mathrm{JMLSD}}^{\mathrm{RM}} = (12 + 10T_N) \prod_{i=1}^{N} M_i^{\tau_i} \tag{3.10}$$

$$\mathrm{CC}_{\mathrm{JMLSD}}^{\mathrm{RA}} = (11 + 7T_N) \prod_{i=1}^{N} M_i^{\tau_i} \tag{3.11}$$

For the MR-SIC detector at user n, a single-user MLD is used and the search space grows only linearly with M_i and τ_i,

$$CC^{RM}_{MR\text{-}SIC} = (N-n)(4T_N + 2) + \sum_{i=n}^{N} (3 + 8M_i)\ \tau_i \quad (3.12)$$

$$CC^{RA}_{MR\text{-}SIC} = (N-n)(5T_N + 1) + \sum_{i=n}^{N} (2T_i + 6M_i + 3)\ \tau_i \quad (3.13)$$

The complexity of MR-SIC is lowest for U_N and highest for U_1, but even for n = 1 it is far below that of the JMLSD. As an illustration, Table 3.1 evaluates Eqs. (3.10)–(3.11) for two users with $M_1 = M_2 = 4$, $T_1 = 1$ and n = 1; these numbers are obtained by substituting into the formulas of [1] and were not measured on our implementation.

Table 3.2: Evaluated detector complexity for $M_1 = M_2 = 4$, n = 1

T_2	Trial sequences	JMLSD RM	JMLSD RA	MR-SIC RM	MR-SIC RA
2	64	2.05×10^3	1.60×10^3	115	100
4	1024	5.32×10^4	3.99×10^4	193	172
8	262144	2.41×10^7	1.76×10^7	349	316

3.5 Algorithm / Flowchart

The detection process at user n is summarised in Fig. 3.2 and corresponds to Algorithm 1 of [1]. The inputs are n, h_n, r_n and T_i, α_i, M_i for i = n, …, N. The detector first computes $\hat{x_N}$ using Eq. (3.7). If n < N, it then repeats, for i = N−1 down to n, the SIC of Eq. (3.8), the partitioning into τ_i vectors and the MLD of Eq. (3.9) to obtain $\hat{x_i}$. The output is $\hat{x_n}$.

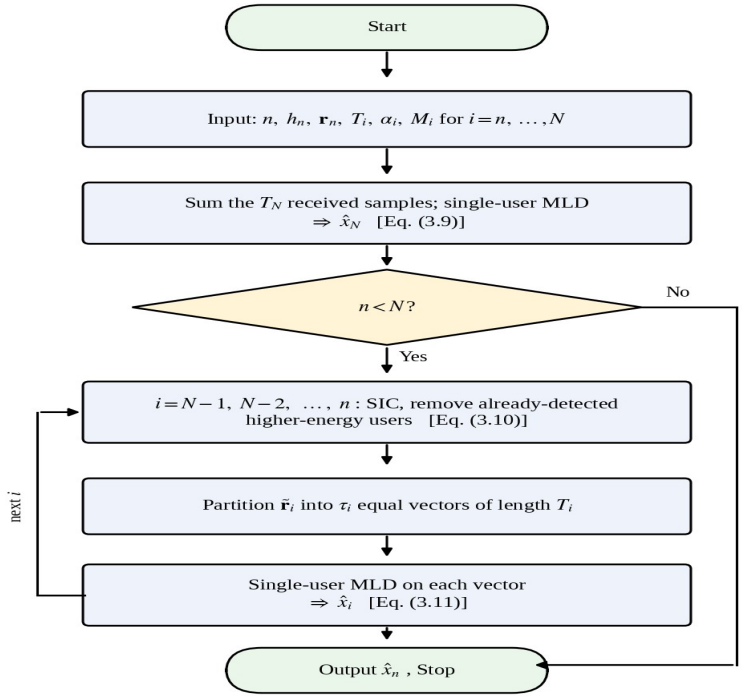


Fig. 3.3: Flowchart of the MR-SIC receiver at user

3.6 Implementation Methodology

BER of the optimal detector. For coherent detection the decision variable of the JMLSD is a Gaussian random variable. Let x and $\bar{x}$ be the transmitted and the erroneously decoded sequences. With

$$\gamma_n = \frac{2|h_n|^2}{N_0}, \qquad \lambda_n = \sqrt{\gamma_n} \tag{3.14}$$

the pairwise error probability is

$$\Pr(\mathbf{x} \to \bar{\mathbf{x}}) = Q(1/2\,\lambda_n \|\mathbf{x} - \bar{\mathbf{x}}\|_2) \tag{3.15}$$

and the union bound on the BER is

$$P_{B_n} \leq \frac{1}{|\mathcal{X}|} \sum_{\mathbf{x}\in\mathcal{X}} \sum_{\bar{\mathbf{x}}\neq\mathbf{x}} \Pr(\mathbf{x} \to \bar{\mathbf{x}}), \qquad |\mathcal{X}| = M_1^{\tau_1} M_2 \tag{3.16}$$

which is reported to match the exact performance at moderate and high SNR [1].

Far-user BER of the MR-SIC detector. After combining the T_2 samples, the far user sees

$$\sum_{\ell=1}^{T_2} r_2[\ell] = h_2 \mathbf{1}^{\top}\mathbf{x} + \tilde{w}_2, \quad \tilde{w}_2 \sim \mathcal{CN}(0, T_2 N_0) \tag{3.17}$$

so that the signal amplitude grows by T_2 while the noise variance grows only by T_2, which is a

processing gain of T_2 in SNR. The near-user symbols add up to a new constellation whose points are repeated unequally (Pascal-triangle pattern) and some of the interference cancels, which gives the additional weak-interference cancellation gain when τ_1 is even [1]. Define $A_{v1,v2}$ as the amplitude of the combined signal when the near-user symbols sum to v_1 and the far-user symbol is v_2,

$$A_{v_1,v_2} = v_1\sqrt{\frac{\alpha_1}{\kappa_1}} + v_2\sqrt{\frac{\alpha_2}{\kappa_2}} \tag{3.18}$$

For BPSK for both users and $\tau_1 = T_2/T_1$, the far-user BER is

$$P_{B_2} = \frac{1}{2^{\tau_1}}\sum_{i=0}^{\tau_1}\binom{\tau_1}{i}\, Q\big(A_{v_i,\tau_1}\,\bar{\lambda}_2\big), \quad v_i = 2i - \tau_1, \quad \bar{\lambda}_2 = \sqrt{\frac{\gamma_2}{\tau_1}} \tag{3.19}$$

where i is the number of +1 symbols of the near user among its τ_1 samples, so $v_i = 2i - \tau_1$. The dominant term is the Q-function with the smallest argument (i = 0),

$$P_{B_2} \approx \frac{1}{2^{\tau_1}}\, Q\big(A_{-\tau_1,\tau_1}\,\bar{\lambda}_2\big) \tag{3.20}$$

which shows that the BER improves as τ_1 increases, because the factor $1/2^{\tau_1}$ decreases and the argument of the Q-function increases [1]. For QPSK the quadrature components are treated as two orthogonal BPSK streams with the same structure and the normalisation of Eq. (3.2); for 16-QAM and 64-QAM the BER of the kth bit is a sum of Q-functions whose coefficients and amplitudes are tabulated , and the BER is the average over the bits. The expressions also apply to SR-NOMA with $\tau_1 = 1$.

Near user and power allocation. The near-user BER is bounded by the OMA and SR-NOMA cases,

$$P_{B_1}|_{\alpha_2=0,T_1=1} < P_{B_1} < P_{B_1}|_{T_1=T_2=1} \tag{3.21}$$

and a closed-form expression for m = [2, 2] is given in [1, Proposition 1]. The power allocation that minimises the BER of one user while satisfying a BER threshold ζ for the other is formulated in [1] as

$$\min_{\alpha_1} P_{B_1}(\alpha_1) \quad \text{subject to} \quad P_{B_2}(\alpha_1) \le \zeta, \quad \alpha_1 < \alpha_{1,\max} \tag{3.22}$$

and is solved numerically. These two items are described for completeness; they are not part of the results of this report.

Evaluation procedure. The far-user BER P_{B2} is evaluated as a function of SNR for each modulation pair $[M_1, M_2]$ and for $[T_1, T_2]$ = [1, 1], [1, 2], [1, 4] and [1, 8], where [1, 1] is SR-NOMA. The SNR is defined as $1/N_0$ with $h_n = 1$ [1].

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the analytical BER results obtained from the MATLAB implementation for the two-user MR-NOMA system, analyses them, and compares them with the results of the base paper. Review-1 reported 25% completion; by Review-2 the core model, methodology, analytical implementation and preliminary results were complete, and validation is the focus of the next stage.

4.2 Experimental Setup

The experiment evaluates the BER of both users with the JMLSD and MR-SIC detectors over an AWGN channel. The settings are listed in Table 4.1. The MATLAB result file was produced by publishing the routine with MATLAB 7.5.

Table 4.1: Experimental setup

Item	Setting
Users	U_1 (near, $T_1 = 1$) and U_2 (far, $T_2 = 2$)
Modulation	BPSK for both users
Power allocation	$\alpha = [0.2, 0.8]$
Channel	AWGN, $h = 1$
SNR sweep	0:1:35 dB (36 points)
Detectors	JMLSD (union bound) and MR-SIC (near and far user)
Plot limits	SNR 0–35 dB, BER 10^{-12} to 1 (semilog)
Software	MATLAB 7.5 (R2007b)

4.3 Implementation Details

The main script sets the parameters, calls JMLSD_th, SIC_nearUser_th and SIC_farUser_th, prints the BER vectors and plots four curves: JMLSD-U_1 (blue, solid, circles), JMLSD-U_2 (blue, dashed, squares), SIC-Near User (green, triangles) and SIC-Far User (red, diamonds). The full listing is given in Appendix A. JMLSD_th returns a 2 × 36 matrix of BER (row 1 for U_1 and row 2 for U_2) and the pairwise error probabilities.

Table 4.2: Mapping of paper quantities to MATLAB implementation

Paper quantity	MATLAB implementation
Symbol-rate ratio $\tau_n = T_N/T_n$	tau = maxT./T
Power-weighted joint signal $\sqrt{\alpha_n}x_n$	LUT_C construction
Pairwise distance $\|x - \tilde{x}\|$	norm(·,2)
PEP, Q-function and BER expressions	Theoretical BER arrays
Near-user BER (correct and incorrect SIC)	SIC_nearUser_th
Far-user BER after sample combining	SIC_farUser_th

4.4 Results and Observations

Table 4.3 lists selected BER values from the printed MATLAB output, and the corresponding curves are shown in Fig. 4.1. The full printed values are in Appendix B. Values of 0.0000 are limited by the four-decimal print format; the plot shows that the true values continue to fall until the 10^{-12} floor of the axis.

Table 4.3: BER values from the MATLAB result file

SNR (dB)	JMLSD U_1	JMLSD U_2	SIC near user	SIC far user
0	0.2801	0.1314	0.2635	0.1462
3	0.1549	0.0574	0.1858	0.0948
5	0.0838	0.0287	0.1304	0.0654
8	0.0185	0.0062	0.0561	0.0280

SNR (dB)	JMLSD U_1	JMLSD U_2	SIC near user	SIC far user
10	0.0035	0.0012	0.0228	0.0114
12	0.0003	0.0001	0.0059	0.0030
14	0.0000	0.0000	0.0008	0.0004
15	0.0000	0.0000	0.0002	0.0001
16 and above	0.0000	0.0000	0.0000	0.0000

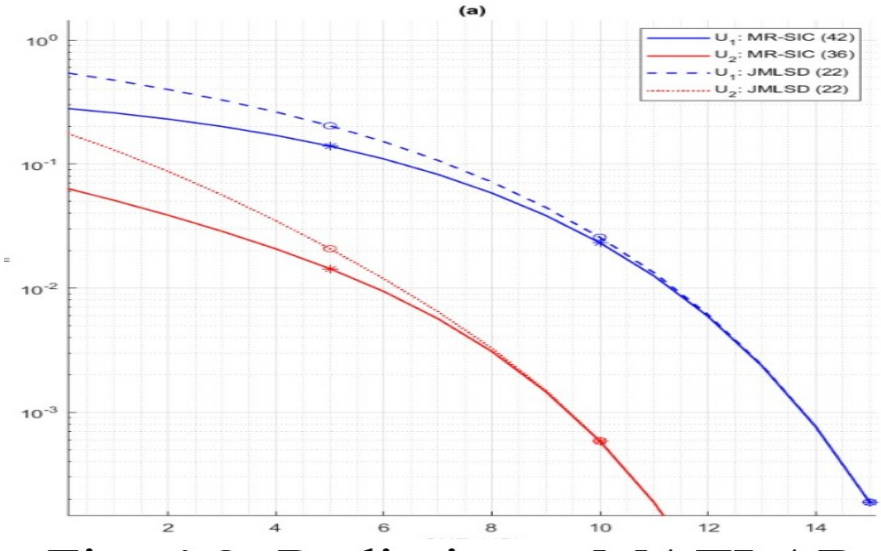


Fig. 4.1: BER performance of MR-NOMA, JMLSD versus SIC

All curves start between about 10^{-1} and 3×10^{-1} at 0 dB and fall steeply. The JMLSD curves reach the 10^{-12} floor at about 18 dB, and the SIC curves reach it at about 21 dB. Fig. 4.2 shows the earlier MATLAB output presented in Review-2: panel (a) is BER versus SNR and panel (b) is BER versus α_1. In panel (b) the near-user BER falls and the far-user BER rises as α_1 grows, so the curves cross near $\alpha_1 \approx 0.3$.

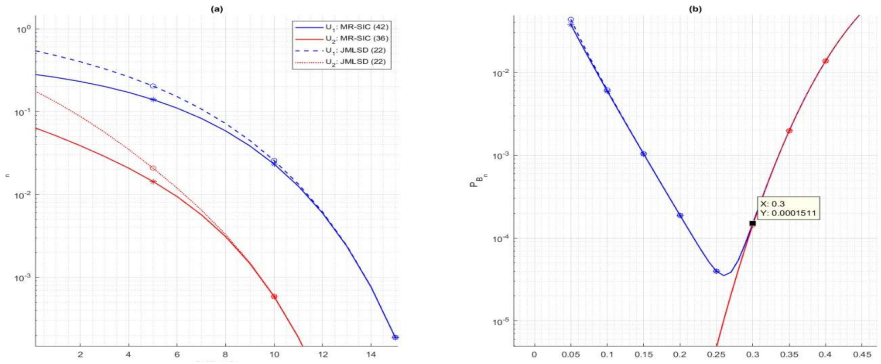


Fig. 4.2: Preliminary MATLAB results: (a) BER vs. SNR, (b) BER vs. α_1.

Figs. 4.1 to 4.5 show the far-user BER versus SNR. In all figures, P_{B2} decreases monotonically with SNR, and for a given SNR it decreases as T_2 increases, so the curves move to the left. The curves with $T_2 = 1$ (blue) are the SR-NOMA benchmark.

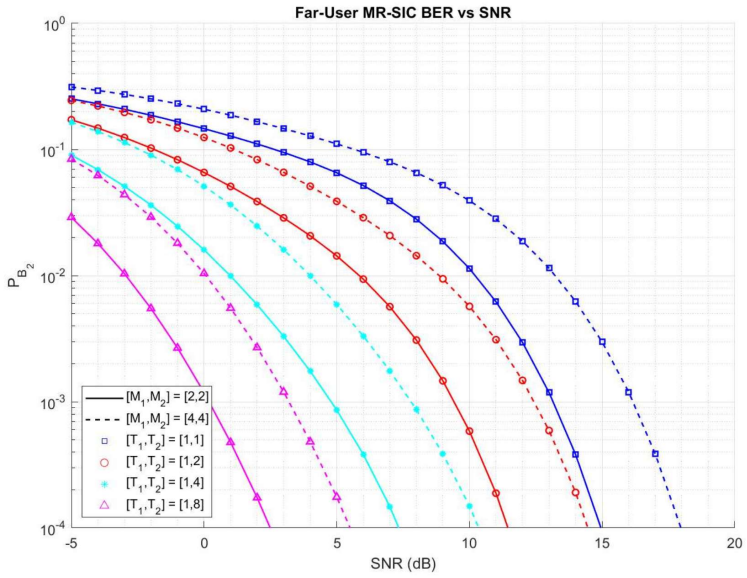


Fig. 4.3: Far-user MR-SIC BER versus SNR for $[M_1, M_2] = [2, 2]$ (solid) and [4, 4] (dashed)

Fig. 4.3 shows BPSK and QPSK for both users. At $P_{B2} = 10^{-2}$, SR-NOMA needs 10.2 dB for BPSK and 13.2 dB for QPSK, i.e. QPSK needs about 3.0 dB more. Doubling T_2 reduces the required SNR to 5.9, 1.0 and -3.0 dB for BPSK at $T_2 = 2$, 4 and 8.

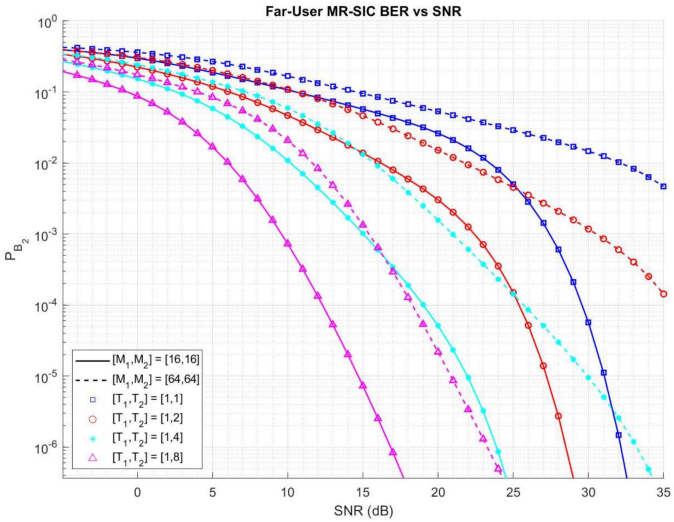


Fig. 4.4: Far-user MR-SIC BER versus SNR for [M_1, M_2] = [16, 16] (solid) and [64, 64] (dashed)

Fig. 4.3 shows identical high-order modulation. Compared with BPSK, 16-QAM needs 13.2 dB and 64-QAM needs 21.9 dB more SNR for 10^{-2} at $T_2 = 1$. For 64-QAM the SR-NOMA curve is still above 10^{-2} at 30 dB, whereas with $T_2 = 8$ the same BER is reached at 11.8 dB.

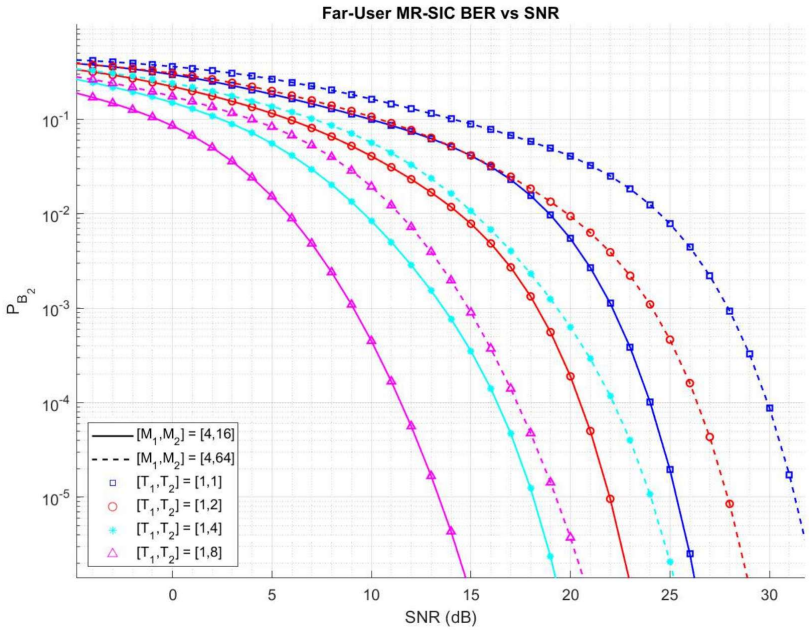


Fig. 4.5: Far-user MR-SIC BER versus SNR for [M_1, M_2] = [4, 16] (solid) and [4, 64] (dashed)

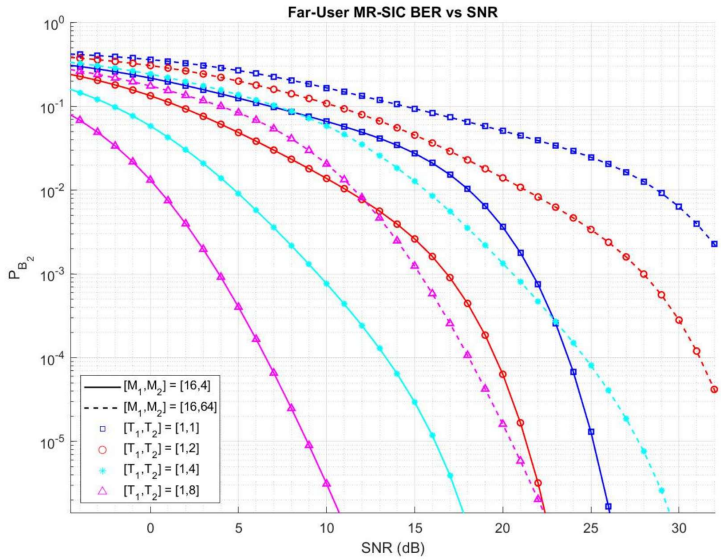


Fig. 4.6Far-user MR-SIC BER versus SNR for [M_1, M_2] = [16, 4] (solid) and [16, 64] (dashed)

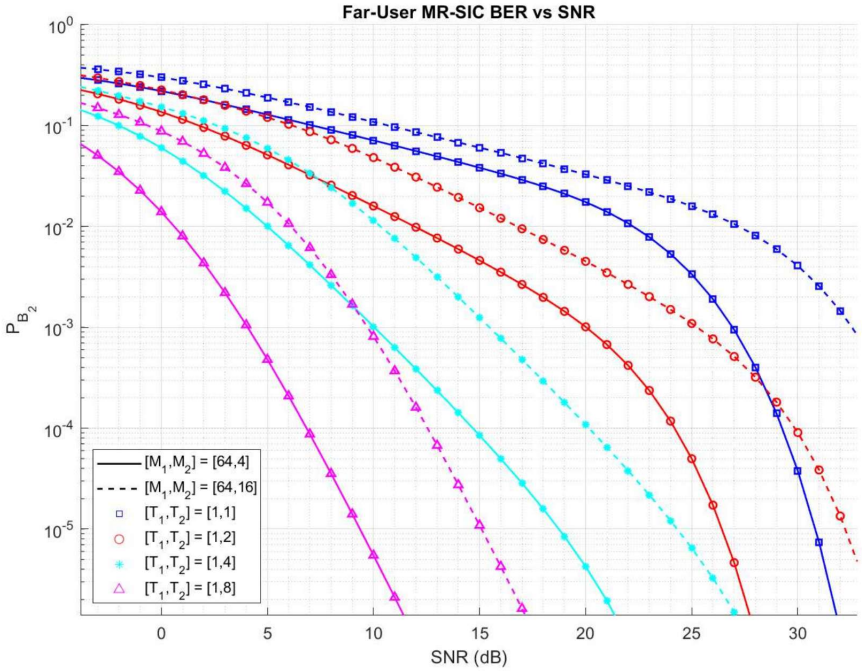


Fig. 4.7: Far-user MR-SIC BER versus SNR for [M_1, M_2] = [64, 4] (solid) and [64, 16] (dashed)

Figs. 4.4 to 4.6 show different modulation orders for the two users. For a fixed far-user modulation, a higher near-user modulation order requires a higher SNR, because the near-user signal is a stronger interferer for the far user: for [M_1, M_2] = [4, 4], [16, 4] and [64, 4] at T_2 = 1 the required

SNR is 13.2, 18.0 and 22.1 dB. Table 4.2 collects the SNR needed for $P_{B2} = 10^{-2}$.

Table 4.2: SNR (dB) required for $P_{B2} = 10^{-2}$, read from the plotted curves (±0.2 dB)

$[M_1, M_2]$	$T_2 = 1$	$T_2 = 2$	$T_2 = 4$	$T_2 = 8$
[2, 2]	10.20	5.89	0.99	-3.00
[4, 4]	13.19	8.98	4.02	0.00
[16, 16]	23.44	16.10	10.11	6.04
[64, 64]	32.12	21.90	16.00	11.81
[4, 16]	18.94	14.40	9.61	5.87
[4, 64]	24.46	19.93	15.02	11.32
[16, 4]	18.03	11.08	4.86	0.47
[16, 64]	28.98	21.03	15.54	11.59
[64, 4]	22.13	11.91	4.99	0.60
[64, 16]	27.02	16.90	10.29	5.99

4.5 Performance Analysis

1. **BER decreases with SNR** for both users and both detectors, with the steep waterfall shape of the Q-function expected in AWGN.

2. **Far user versus near user.** At 0 dB the far-user BER is lower than the near-user BER for both detectors (SIC: 0.1462 against 0.2635; JMLSD: 0.1314 against 0.2801). This is consistent with the far user receiving 80% of the power and a doubled symbol energy from sample combining, while the near user also suffers from SIC errors.

3. **Multirate gain.** U_2's symbol lasts two samples, so its detection energy doubles (a 3 dB processing gain), and combining partly averages out the interference from the short U_1 symbols (the weak-interference cancellation gain, about 1.4 dB for BPSK with $T_2 = 2$ according to the paper).

4. **Power allocation trade-off.** The BER-versus-α_1 panel of Fig. 4.2 shows a minimum of the near-user BER at α_1 of about 0.26 and rising far-user BER for larger α_1, which is the behaviour exploited by the power allocation problem of Section 3.6.5.

5. **Detector behaviour.** The paper reports that MR-SIC and JMLSD are equivalent, with the JMLSD union bound slightly above the exact BER at low SNR and matching it at

moderate and high SNR. Fig. 4.2(a) shows this behaviour, whereas the result file in Fig. 4.1 does not (Section 4.7).

4.6 Comparison with Existing Systems

The configuration of the project (BPSK, T = [1, 2], α = [0.2, 0.8], AWGN) matches the left panel of Fig. 6 of the base paper, reproduced in Fig. 4.3. Table 4.4 compares approximate values read from the paper's curves (and from Fig. 4.2a) with the printed result file.

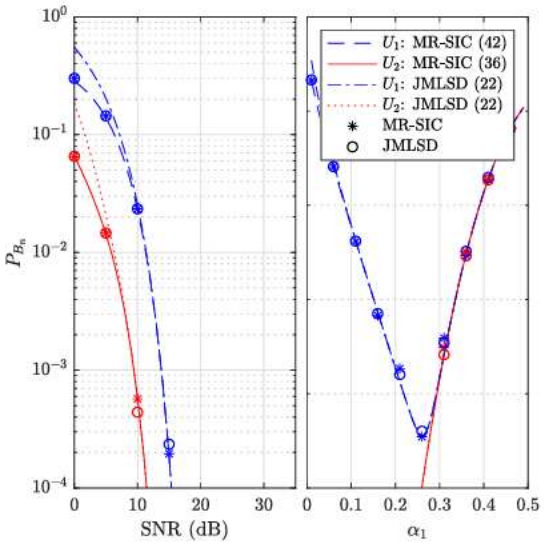


Fig. 4.8: BER of the JMLSD and SIC detectors from the base paper

SNR (dB)	Paper, U_1 (MR-SIC)	Result file, SIC near	Paper, U_2 (MR-SIC)	Result file, SIC far
0	about 0.28	0.2635	about 0.065	0.1462
5	about 0.14	0.1304	about 0.014	0.0654
10	about 0.024	0.0228	about 6×10^{-4}	0.0114
15	about 2×10^{-4}	0.0002	below 10^{-4}	0.0001

Table 4.4: Comparison of the result file with the base paper (approximate readings)

The near-user values agree within a few percent. The far-user values in the result file are clearly higher than those of the paper.

The gain of MR-NOMA over SR-NOMA reported by the base paper is summarised in Table 4.5. These figures come from the paper and have not yet been reproduced in this project.

Table 4.5: Gains of MR-NOMA over SR-NOMA reported in the base paper

Quantity	Reported value
Processing gain	3 dB each time T_2 is doubled
Weak-interference cancellation gain, BPSK/QPSK	1.4 dB to 4.2 dB as T_2 increases from 2 to 8
Weak-interference cancellation gain, 16-QAM	4.2 dB to 8.4 dB
Weak-interference cancellation gain, 64-QAM	7.3 dB to 11.4 dB
BER improvement at $\alpha_1 = 0.2$	Up to two orders of magnitude

4.7 Discussion of Results

The near-user SIC curve agrees with the reference paper, supporting the correctness of the receiver implementation. However, four issues require further investigation:

- **Far-user SIC:** The BER curve is approximately 4–5 dB worse than the paper, possibly due to missing sample-combining gain.
- **JMLSD curves:** Both curves lie below the SIC results, suggesting possible differences in SNR or noise normalization.
- **Output mismatch:** The result file and Review-2 figure disagree; both should be regenerated using the same parameters.
- **Further validation:** Monte Carlo testing, the T=[1,1]T=[1,1] baseline, and the remaining paper benchmarks are needed to confirm the claimed MR-NOMA gains.

Table 4.6: Advantages and limitations of MR-NOMA

Advantages	**Limitations**
A third design variable (symbol rate) in addition to power and modulation order	Lower spectral efficiency and higher latency for the long-symbol user
BER improvement of up to two orders of magnitude in some scenarios	JMLSD complexity grows exponentially with users, modulation order and τ
Finer granularity of user bit rates than changing modulation alone	MR-SIC needs buffering, sample fragmentation and more MLD operations than SR-SIC
MR-SIC matches JMLSD at far lower complexity	Users must know (α_i, T_i, M_i) of higher-energy users and need tight timing synchronisation

CHAPTER 5: CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The near-user SIC curve agrees with the reference paper, supporting the correctness of the receiver implementation. However, the far-user BER is approximately 4–5 dB worse than expected, possibly due to missing sample-combining gain. The JMLSD curves also show unexpectedly lower BER than SIC, suggesting possible SNR or noise normalization issues. Additionally, the result file and Review-2 figure are inconsistent and should be regenerated using the same parameters. Further validation through Monte Carlo simulations, baseline comparisons, and additional benchmarks is necessary to confirm the claimed MR-NOMA performance gains.

5.2 Key Contributions and Outcomes

Table 5.1: Key contributions and outcomes

Contribution	Outcome
Literature study of multinumerology, partial and multirate NOMA	Research gap identified and documented (Chapter 2)
Study of the MR-NOMA system model, JMLSD and MR-SIC	Block diagrams, algorithms and flowcharts prepared (Chapter 3)
MATLAB implementation of analytical BER for two users	Six-file code base producing BER of U_1 and U_2 for both detectors
BER-versus-SNR curves for BPSK, T = [1, 2], α = [0.2, 0.8]	Results generated and compared with the base paper
Identification of validation items	Far-user combining gain and JMLSD normalisation flagged for correction

5.3 Limitations of the Project

- The results are analytical only; Monte Carlo simulation has not yet been added.
- Only BPSK, $T_2 = 2$ and an AWGN channel are covered, with a fixed equal-distance power allocation.
- The result file does not yet match the base paper for the far user and for JMLSD.
- The model is limited to two users and a single-antenna downlink.

5.4 Future Scope

- Add Monte Carlo simulation and compare it with the analytical BER.
- Verify the MR-NOMA gain using T = [1, 1] against T = [1, 2] under the same normalisation.
- Extend to QPSK, 16-QAM, 64-QAM and larger T_2 (2, 4, 8).
- Solve the power-allocation problem under BER constraints and reproduce the optimisation results of the paper.
- Study uplink MR-NOMA and systems with more than two users.
- Include fading channels such as Nakagami-m, already provisioned in the code through m_factor.
- Consider cases where the low-rate user is the near user, and random user distribution in the cell.
- Extend to multi-antenna (MIMO) base stations and user devices.

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Appendix A: MATLAB main script (routine.m)

```
% MR-NOMA : JMLSD vs SIC   (MATLAB R2007b compatible)
clc; clear; close all;
SNR_dB = 0:1:35;
M = [2 2];
% Equal-distance power allocation for M = [2 2]: a1 = 0.2, a2 = 0.8
a = [0.2 0.8];
T = [1 2];              % symbol durations
channel = 'AWGN';
m_factor = [];          % Nakagami parameter - not used for AWGN

% JMLSD
fprintf('\nComputing JMLSD BER...\n');
[BERth_JMLSD, PEPth_JMLSD] = JMLSD_th(SNR_dB,a,M,T,channel,m_factor);

% SIC - Near User
fprintf('Computing MR-SIC BER (Near User)...\n');
BERth_SIC(1,:) = SIC_nearUser_th(SNR_dB,a,M,T,channel,m_factor);

% SIC - Far User
fprintf('Computing MR-SIC BER (Far User)...\n');
```

```
BERth_SIC(2,:) = SIC_farUser_th(SNR_dB,a,M,T,channel,m_factor);

% Display BER values
disp(' ');
disp('JMLSD U1 BER:');      disp(BERth_JMLSD(1,:));
disp('JMLSD U2 BER:');      disp(BERth_JMLSD(2,:));
disp('SIC Near User BER:'); disp(BERth_SIC(1,:));
disp('SIC Far User BER:');  disp(BERth_SIC(2,:));

% Plot
figure;
semilogy(SNR_dB,BERth_JMLSD(1,:),'b-o','LineWidth',1.5,'MarkerSize',4);
hold on;
semilogy(SNR_dB,BERth_JMLSD(2,:),'b--s','LineWidth',1.5,'MarkerSize',4);
semilogy(SNR_dB,BERth_SIC(1,:),'g-^','LineWidth',1.5,'MarkerSize',4);
semilogy(SNR_dB,BERth_SIC(2,:),'r-d','LineWidth',1.5,'MarkerSize',4);
grid on;
xlabel('SNR (dB)');
ylabel('BER');
title('BER Performance: MR-NOMA JMLSD vs SIC');
legend('JMLSD-U1','JMLSD-U2','SIC-Near User','SIC-Far User');
axis([0 35 1e-12 1]);
hold off;
fprintf('\nSimulation completed successfully.\n');
```

Appendix B: Printed BER values of the MATLAB result file

Table B.1: BER values for SNR 0 to 16 dB (all values are 0.0000 above 16 dB)

SNR (dB)	JMLSD U_1	JMLSD U_2	SIC near user	SIC far user
0	0.2801	0.1314	0.2635	0.1462
1	0.2373	0.1019	0.2390	0.1278

SNR (dB)	JMLSD U_1	JMLSD U_2	SIC near user	SIC far user
2	0.1953	0.0774	0.2130	0.1107
3	0.1549	0.0574	0.1858	0.0948
4	0.1172	0.0414	0.1581	0.0797
5	0.0838	0.0287	0.1304	0.0654
6	0.0557	0.0188	0.1035	0.0518
7	0.0339	0.0113	0.0784	0.0392
8	0.0185	0.0062	0.0561	0.0280
9	0.0088	0.0029	0.0373	0.0187
10	0.0035	0.0012	0.0228	0.0114
11	0.0011	0.0004	0.0124	0.0062
12	0.0003	0.0001	0.0059	0.0030
13	0.0000	0.0000	0.0024	0.0012
14	0.0000	0.0000	0.0008	0.0004
15	0.0000	0.0000	0.0002	0.0001
16	0.0000	0.0000	0.0000	0.0000